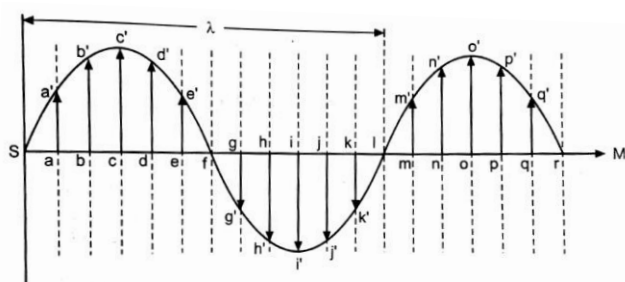


Unit 3 - Oscillation and waves

(M2 proton)

Periodic motion – time period, frequency, displacement as a function of time, periodic functions. Simple harmonic motion (S.H.M) and its equation; phase; oscillations of a loaded spring restoring force and force constant; energy in S.H.M. Kinetic and potential energies;

Representation of a wave:



Few Definitions:

- **WAVE:** A wave is a disturbance that transfers energy through matter or space.
 - **WAVE MOTION:** Wave motion is a form of disturbance which travels through a medium due to the repeated periodic motion of the particles of the medium about their mean positions, the motion being handed on from one particle to next after regular interval of time.
 - **WAVE VELOCITY (v):** Wave velocity is defined as the distance covered by the disturbance (wave) in one second in a given direction.
 - **WAVE PERIOD (T):** The time taken by a particle to complete one oscillation in the wave.
 - **WAVE FREQUENCY (n):** The number of oscillations completed by a particle in one second is called frequency.
 - **WAVELENGTH (λ):** The distance between two consecutive particles of the medium which are in the same phase is called wavelength.
 - **AMPLITUDE OF WAVE (A):** The maximum displacement of the particle from its mean position on either side is called Amplitude of the wave.
 - **VIBRATORY MOTION:** The back and forth motion of the particle (or body) about its equilibrium position is called vibratory motion.
 - **PERIODIC MOTION:** A motion which repeats again and again after a regular interval of time is called a periodic motion.
-

Relation between Velocity, Frequency and wavelength

- Velocity (v) is the distance travelled by the wave in one second.
- Period (T) is the time required to complete one oscillation.
- Frequency (f) is the number of oscillations completed in one second.
- Wavelength (λ) is the length of one full wave.

Hence,
$$\text{velocity} = \frac{\text{Distance covered}}{\text{time taken}}$$

When disturbance travels through one full wave, then,

$$\text{Distance covered} = \text{Wavelength} = \lambda$$

$$\text{Time taken} = \frac{\text{Wavelength}}{\text{Period}}$$

$$v = \frac{\lambda}{T} \quad \text{and} \quad \frac{1}{T} = f$$

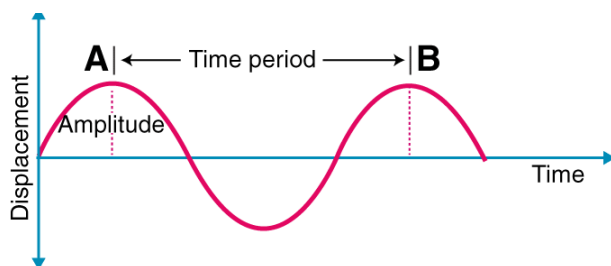
$$\therefore v = f \lambda$$

Periodic motion: Time period & Frequency

A motion that repeats itself after equal intervals of time is known as periodic motion.

Example:

- On analysing the motion of tuning fork or motion of a pendulum it is observed that the pendulum passes through the mean position only after a definite interval of time.
- We can also classify the above motion to be oscillatory.
- An oscillatory is a motion in which the body moves to and fro about a fixed position. So an oscillatory motion can be periodic but it is not necessary. So taking an example of a wave motion we will see some parameters related to periodic motion. Consider the following figure:



Time Period (T): It is the time taken by the motion to repeat itself. So the unit of a time period is seconds.

Frequency (f): It is defined as a number of times the motion is repeated in one second. The unit of frequency is Hz (Hertz). Frequency is related to Time period as,

$$f = \frac{1}{T}$$

Displacement as a function of time, periodic functions

To understand this idea of displacement as a function of time, we will have to derive an expression for displacement, assume a body traveling at an initial velocity of v_1 at the time t_1 and then the body accelerates at a constant acceleration of 'a' for some time and a final velocity of v_2 at the time t_2 , keeping these things in assumption let's derive the following.



Let's write displacement as

$$d = V_{average} * \Delta t$$

Where Δt is the change in time, assuming that the object is under constant acceleration.

$$d = \left(\frac{V_1 + V_2}{2} \right) * \Delta t$$

Where V_2 and V_1 are final and initial velocities respectively, let's rewrite final velocity in terms of initial velocity for the sake of simplicity.

$$d = \left(\frac{V_1 + (V_1 + a * \Delta t)}{2} \right) * \Delta t$$

Where a is the constant acceleration the body is moving at, now if we rewrite the above as,

$$d = \left(\frac{2 * V_1}{2} + \frac{a * \Delta t}{2} \right) * \Delta t$$

The above expression is one of the most fundamental expressions in kinematics, it is also sometimes given as

$$d = V_i t + \frac{1}{2} a t^2$$

Where V_i is the initial velocity, and t is actually the change in time, all the quantities in this derivation, like Velocity, displacement and acceleration, are vector quantities.

If we have a look at our final expression, it is very clear that change in displacement depends upon time.

Simple Harmonic, Periodic and Oscillation Motion

Simple harmonic motion can be described as an oscillatory motion in which the acceleration of the particle at any position is directly proportional to the displacement from the mean position. It is a special case of oscillatory motion.

All the Simple Harmonic Motions are oscillatory and also periodic, but not all oscillatory motions are SHM. Oscillatory motion is also called the harmonic motion of all the oscillatory motions, wherein the most important one is simple harmonic motion (SHM).

In this type of oscillatory motion, displacement, velocity and acceleration, and force vary (w.r.t time) in a way that can be described by either sine (or) the cosine functions collectively called sinusoids.

The study of Simple Harmonic Motion is very useful and forms an important tool in understanding the characteristics of sound waves, light waves and alternating currents.

Difference between Periodic, Oscillation and Simple Harmonic Motion

Periodic Motion

- A motion repeats itself after an equal interval of time. **For example, uniform circular motion.**
- There is no equilibrium position.
- There is no restoring force.
- There is no stable equilibrium position.

Oscillation Motion

- To and fro motion of a particle about a mean position is called an oscillatory motion in which a particle moves on either side of the equilibrium (or) mean position is an oscillatory motion.
- It is a kind of periodic motion bounded between two extreme points. **For example,** Oscillation of Simple Pendulum, Spring-Mass System.
- The object will keep on moving between two extreme points about a fixed point is called the mean position (or) equilibrium position along any path. (the path is not a constraint).
- There will be a restoring force directed towards equilibrium position (or) mean position.
- **In an oscillatory motion,** the net force on the particle is zero at the mean position.
- The mean position is a stable equilibrium position.

Simple Harmonic Motion or SHM

- It is a special case of oscillation along with a straight line between the two extreme points (the path of SHM is a constraint).
- Path of the object needs to be a straight line.
- There will be a restoring force directed towards the equilibrium position (or) mean position.
- Mean position in Simple harmonic motion is a stable equilibrium.

Conditions for SHM:

→

$$F \propto -x$$

→

$$A \propto -x$$

Types of Simple Harmonic Motion

SHM or Simple Harmonic Motion can be classified into two types:

- Linear SHM
- Angular SHM

Linear Simple Harmonic Motion

When a particle moves to and fro about a fixed point (called equilibrium position) along with a straight line then its motion is called linear Simple Harmonic Motion.

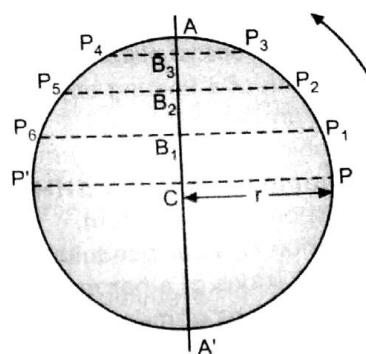
Angular Simple Harmonic Motion

When a system oscillates angular long with respect to a fixed axis, then its motion is called angular simple harmonic motion.

LINEAR SIMPLE HARMONIC MOTION

Linear S.H.M. is defined as the periodic motion of a body along a straight line in which force is always directed towards mean position and its magnitude is proportional to the displacement from mean position.

- Consider a point P moving along a circle in anticlockwise direction with constant speed.
- Let r be the radius of the circular path and C be the centre.
- The perpendicular from P on diameter AA' meets at centre C.
- C is called the projection of P on diameter.
- Also, B₁, B₂, B₃ are the projections of particles at positions P₁, P₂, P₃.
- Thus, when a particle moves from P to A, its projection moves from C to A.
- When particle completes one rotation along circle, its projection will complete one oscillation along a straight line.
- Hence, projection of a particle is moving up and down and always tends to move towards centre C (mean position) i.e. performing linear S.H.M.
- *The maximum possible displacement of projection is called amplitude 'a' and is equal to radius r.*
- S.H.M. can also be defined as the projection of a uniform circular motion on diameter of a circle.



CHARACTERISTICS OF LINEAR S.H.M.

- 1) Motion is periodic along straight line.
- 2) Force is directed towards the mean position
- 3) Force is directly proportional to its displacement from mean position.
- 4) Velocity of particle is maximum at the centre and minimum at extreme position.

Simple Harmonic Motion Key Terms

Mean Position

The point at which net force acting on the particle is zero.

From the mean position, the force acting on the particle is,

$$\vec{F} \propto -\vec{x}$$

$$\vec{a} \propto -\vec{x}$$

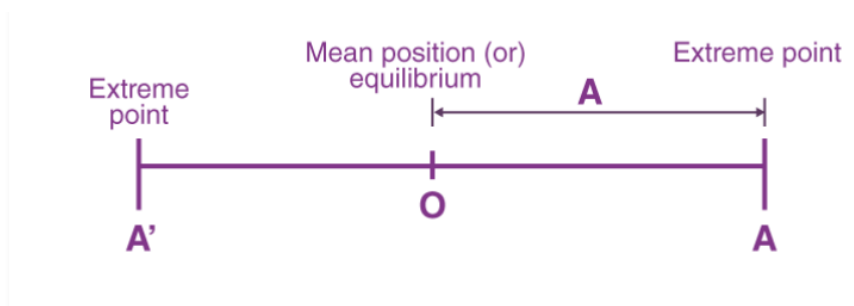
Conditions at Mean Position

$$\vec{F}_{net} = 0$$

$$\vec{a} = 0$$

The force acting on the particle is negative of the displacement. So, this point of equilibrium will be a stable equilibrium.

Amplitude in SHM: The maximum displacement of the particle from the mean position.



Time Period and Frequency of SHM

The minimum time after which the particle keeps on repeating its motion is known as the time period (or) the shortest time taken to complete one oscillation is also defined as the time period.

$$T = 2\pi/\omega$$

Frequency: The number of oscillations per second is defined as the **frequency**.

$$\text{Frequency} = 1/T \text{ and, angular frequency } \omega = 2\pi f = 2\pi/T$$

Phase in SHM

The phase of a vibrating particle at any instant is the state of the vibrating (or) oscillating particle regarding its displacement and direction of vibration at that particular instant.

The expression, and position of a particle as a function of time.

$$x = A \sin (\omega t + \Phi)$$

Where $(\omega t + \Phi)$ is the phase of the particle, the **phase angle** at time $t = 0$ is known as the initial phase.

Phase Difference

The difference of total phase angles of two particles executing simple harmonic motion with respect to the mean position is known as the phase difference. Two vibrating particles are said to be in the same phase; the phase difference between them is an even multiple of π .

$$\Delta\Phi = n\pi \text{ where } n = 0, 1, 2, 3, \dots$$

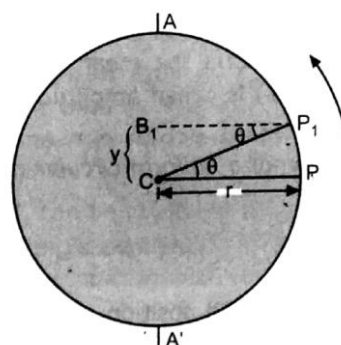
Two vibrating particles are said to be in the opposite phases if the phase difference between them is an odd multiple of π .

$$\Delta\Phi = (2n + 1) \pi \text{ where } n = 0, 1, 2, 3, \dots$$

EQUATION OF S.H.M.

1) Expression for Displacement in S.H.M.

- Consider the particle P performing uniform circular motion in anticlockwise direction.
- Let, ω be the angular velocity of the particle in circular motion.
- r be the radius of circular path called radius vector.
- C the centre of circular path



(I) Particle starts from the mean position:

- The particle is initially at position P, and C is its projection.
- The particle moves from P to P1 in time t seconds.
- Its projection moves from C to B1
- Hence, $CB1 = y$ is the displacement of projection in time t seconds.
- Let θ be the angular displacement of particle in time t seconds.

From the figure,

$$\text{In } \triangle CP1B1, \quad \sin \theta = \frac{CB1}{CP1} \quad \text{i.e. } \sin \theta = \frac{y}{r}$$

$$\therefore y = r \sin \theta$$

$$\text{But } \theta = \omega t \text{ hence, } y = r \sin \omega t$$

This is the equation of S.H.M.

- r is the maximum possible displacement of the particle in S.H.M. called amplitude 'A'

$$\text{With } r = a, \quad y = A \sin \omega t$$

(II) General case: Particle is at any position between mean and extreme:

- Consider particle is initially at position P, radius is making an angle α with the horizontal as shown in the figure.
- The particle moves from P to P1 in time t seconds.
- Its projection is at position B1.
- $CB1 = y$ is the displacement of projection from mean position.

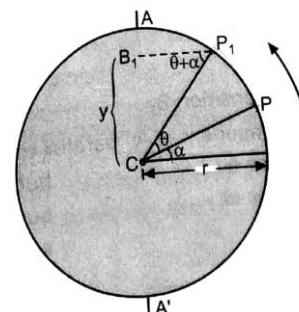
$$\text{In } \triangle CP1B1, \quad \sin(\theta + \alpha) = \frac{CB1}{CP1}$$

$$\text{i.e. } \sin(\theta + \alpha) = \frac{y}{r}$$

$$\therefore y = r \sin(\theta + \alpha)$$

$$\text{But } \theta = \omega t \text{ hence, } y = r \sin(\omega t + \alpha)$$

Or $y = A \sin(\omega t + \alpha)$, where A is the amplitude



- PHASE ANGLE:** The angle which gives position, direction and displacement of the particle in SHM at any instant is known as phase angle.

- **EPOCH:** It is the initial phase angle made by the radius vector with the horizontal.

(III) Particle starts from extreme position:

If the particle starts from extreme position, then $\alpha = 90^\circ$

$$y = r \sin(\omega t + 90^\circ)$$

$$y = r \cos \omega t \text{ or } y = A \cos \omega t$$

2) Expression for Velocity in S.H.M.

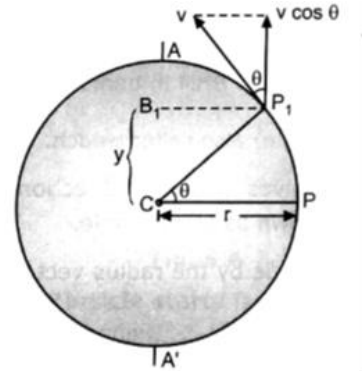
- To find the velocity of particle in SHM, a component of velocity of the reference particle P is to be considered which is parallel to the diameter AA'.
- Let, r – radius of reference circle.

v - linear velocity of reference particle P

θ – angular displacement

CB1 = y = Displacement of particle in SHM

ω - angular velocity



- Velocity of particle in SHM at position B1
 - = component of v parallel to AA'
 - = $v \cos \theta$ (but $\theta = \omega t$)
 - = $v \cos \omega t$ ($\cos \omega t = \sqrt{1 - \sin^2 \omega t}$)
 - And $v = r\omega$ or $v = a\omega$
 - = $A \omega \sqrt{1 - \sin^2 \omega t}$ ($y = A \sin \omega t$ or $y/A = \sin \omega t$)
 - = $A \omega \sqrt{1 - \left(\frac{y}{a}\right)^2}$

- On simplifying this equation, we can get the velocity of particle in SHM as

$$v = \omega \sqrt{A^2 - y^2} \quad \text{where } a \text{ is the amplitude and } y \text{ is the displacement.}$$

CASE 1: When a particle is at mean position, i.e. $y = 0$

$$v = \omega \sqrt{a^2 - 0}$$

$$v_{\max} = A \omega$$

The velocity v is maximum at mean position.

CASE 2: When a particle is at extreme position, i.e. $y = a$,

$$v = \omega \sqrt{A^2 - A^2} \text{ or } v = 0$$

The velocity v is minimum at extreme position.

3) Expression for Acceleration in S.H.M.

- Reference particle performing uniform circular motion has acceleration always directed towards centre and along radius, called radial acceleration.

$$\text{Radial acceleration, } a = r \omega^2$$

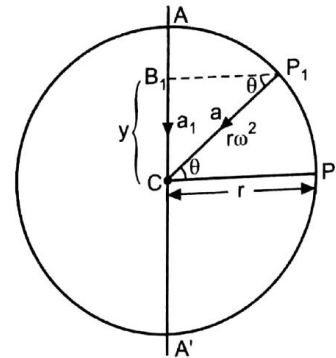
- The component of radial acceleration along parallel to AA' gives acceleration of particle in SHM at position B1
- Let, r – radius of reference circle.

$$\mathbf{a} = r \omega^2 \text{ - radial acceleration}$$

θ – angular displacement

$$CB1 = y = \text{Displacement of particle in SHM}$$

ω - angular velocity



- Acceleration of the particle in SHM
 - = component of radial acceleration along parallel direction of AA'
 - = $a \sin \theta$ ($\theta = \omega t$, and $\mathbf{a} = r \omega^2$)
 - = $r \omega^2 \sin \omega t$
 - But $r \sin \omega t = y$

Hence, the acceleration of particle in SHM is $\mathbf{a} = \omega^2 y$

This acceleration is directed towards the mean position.

CASE 1: When a particle is at mean position, i.e. $y = 0$ $\mathbf{a} = 0$

The acceleration is minimum at mean position

CASE 2: When a particle is at extreme position, i.e. $y = A$, $\mathbf{a} = \omega^2 A$

The acceleration is maximum at extreme position.

Energy in Simple Harmonic Motion (SHM)

The system that executes SHM is called the harmonic oscillator.

Consider a particle of mass m , executing linear simple harmonic motion of **angular frequency** (ω) and amplitude (A),

the displacement $(\vec{x})$, velocity $(\vec{v})$ and acceleration $(\vec{a})$ at any time t are given by

$$x = A \sin(\omega t + \phi)$$

$$v = A\omega \cos(\omega t + \phi) = \omega\sqrt{A^2 - x^2}$$

$$a = -\omega^2 A \sin(\omega t + \phi) = -\omega^2 x$$

The restoring force $(\vec{F})$ acting on the particle is given by

$$F = -kx, \text{ where } k = m\omega^2.$$

Kinetic Energy of a Particle in SHM

Kinetic Energy

$$= \frac{1}{2}mv^2 \text{ [Since, } v^2 = A^2\omega^2\cos^2(\omega t + \phi)\text{]}$$

$$= \frac{1}{2}m\omega^2 A^2\cos^2(\omega t + \phi)$$

$$= \frac{1}{2}m\omega^2 (A^2 - x^2)$$

Therefore, the Kinetic Energy

$$= \frac{1}{2}m\omega^2 A^2\cos^2(\omega t + \phi) = \frac{1}{2}m\omega^2 (A^2 - x^2)$$

Potential Energy of SHM

The total work done by the restoring force in displacing the particle from ($x = 0$) (mean position) to $x = x$:

When the particle has been displaced from x to $x + dx$, the work done by restoring force is

$$dw = F dx = -kx dx$$

$$w = \int dw = \int_0^x -kx dx = -\frac{kx^2}{2}$$

$$= -\frac{m\omega^2 x^2}{2}$$

$$[k = m\omega^2]$$

$$= -\frac{m\omega^2}{2} A^2 \sin^2(\omega t + \phi)$$

Potential Energy = -(work done by restoring force)

$$= \frac{m\omega^2 x^2}{2} = \frac{m\omega^2 A^2}{2} \sin^2(\omega t + \phi)$$

Total Mechanical Energy of the Particle Executing SHM

$$E = KE + PE$$

$$E = \frac{1}{2} m\omega^2 (A^2 - x^2) + \frac{1}{2} m\omega^2 x^2$$

$$E = \frac{1}{2} m\omega^2 A^2$$

Hence, the particle's total energy in SHM is constant, independent of the instantaneous displacement.

⇒ Relationship between Kinetic Energy, Potential Energy and time in Simple Harmonic Motion at $t = 0$, when $x = \pm A$.

